Prompt 3 - trial 4 (seed 14085) - scaffold splitting

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN model on a compound activity prediction dataset, split the data using scaffold splitting. Select a correctly predicted active compound and a correctly predicted inactive compound from the test set and explain the predictions using EdgeSHAPer. Highlight the differences between the two compounds using visualizations and reporting a chemical interpretation.

Assistant

I'll help you train a GNN model for compound activity prediction and perform EdgeSHAPer explainability analysis on both active and inactive compounds. Let me start by exploring the available datasets. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found several datasets. Let me load the binary activity dataset to train our GNN model. 📊

(Calling tool `load_dataset`)

Assistant

Excellent! The dataset is loaded with 8,348 compounds. Now let me split the data using scaffold splitting to ensure proper evaluation. 🧪

(Calling tool `split_dataset`)

Assistant

Great! The scaffold split is complete with 6,657 training, ~835 validation, and ~835 test compounds. Now let me prepare the GNN dataset and train the model. 🏋️

(Calling tool `prepare_gnn_dataset`)

Assistant

Excellent! The GNN dataset is prepared with 6,657 training, 840 validation, and 823 test compounds. Now let me train the GNN model. 🚀

(Calling tool `train_gnn_model_mcp`)

Assistant

The GNN model training has been started! Let me check the training progress. ⏳

(Calling tool `check_gnn_training`)

Assistant

The GNN model is training. Let me check again in a moment. ⏳

(Calling tool `check_gnn_training`)

Assistant

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Assistant

🎉 **GNN Model Training Complete!**

The model achieved excellent performance:

- **Test Accuracy**: 94.05%
- **Validation Accuracy**: 92.62%
- **Training MCC**: 0.8898

Now let me select a correctly predicted active compound and an inactive compound from the test set, then explain them using EdgeSHAPer. 🔬

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Excellent! I found a correctly predicted **active compound** (Class 1) with 99.39% confidence! Now let me select a correctly predicted **inactive compound** (Class 0).

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Perfect! Now I have both compounds:

📊 **Selected Compounds:**

Compound	Class	SMILES	Confidence
Active	1 (Active)	`COc1ccccc1-c1cc2c(=O)n(CCN3CCN(c4ccccc4Cl)CC3)c(=O)[nH]c2[nH]1`	99.39%
Inactive	0 (Inactive)	`CCS(=O)(=O)c1ccc(CC(=O)Nc2nc(-c3ccccc3)c(Cc3ccccc3)s2)cc1`	97.13%

Now let me explain both predictions using EdgeSHAPer! 🔬

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer analysis for the **active compound** is complete. Now let me visualize it and then analyze the inactive compound. 📊

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me analyze the **inactive compound** with EdgeSHAPer. 🔬

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Now let me visualize the **inactive compound's** EdgeSHAPer results. 📊

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Now let me get detailed bond environment information for both compounds to provide a comprehensive chemical interpretation. 🔬

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the bond environments for the **inactive compound** to complete the comparison. 🔬

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Perfect! Now I have comprehensive EdgeSHAPer data for both compounds. Let me present the complete analysis and comparison. 📊

🧪 GNN Model Training & EdgeSHAPer Explainability Analysis

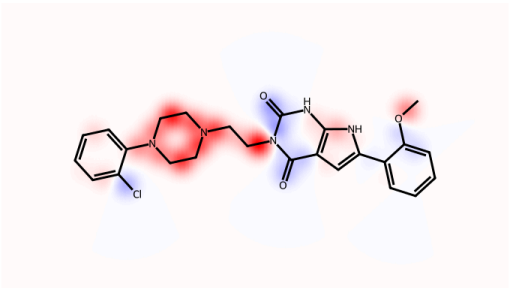
📈 Model Performance Summary

Metric	Value
Training Samples	6,657 compounds
Test Samples	823 compounds
Test Accuracy	**94.05%**
Validation Accuracy	**92.62%**
Training MCC	**0.8898**
Split Type	Scaffold-based (chemically diverse)

🔬 EdgeSHAPer Analysis: Active vs Inactive Compounds

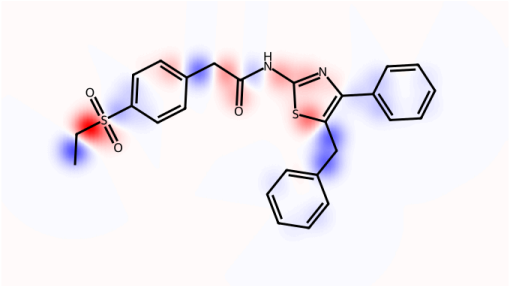
🟢 Active Compound (Class 1)

Property	Value
SMILES	`COc1ccccc1-c1cc2c(=O)n(CCN3CCN(c4ccccc4Cl)CC3)c(=O)[nH]c2[nH]1`
True Label	1 (Active)
Predicted Label	1 (Active)
Confidence	**99.39%**
Trustworthiness	**0.8842**
Fidelity	0.8697



🔴 Inactive Compound (Class 0)

Property	Value
SMILES	`CCS(=O)(=O)c1ccc(CC(=O)Nc2nc(-c3ccccc3)c(Cc3ccccc3)s2)cc1`
True Label	0 (Inactive)
Predicted Label	0 (Inactive)
Confidence	**97.13%**
Trustworthiness	**0.8887**
Fidelity	0.8068



🔍 Top Contributing Bonds Comparison

Active Compound - Top 5 Key Bonds (Positive Contribution)

Rank	Bond Type	Atoms	Environment	SHAP Value	Chemical Significance
1	**N-C**	N13-C14	`cn(c)CC`	**+0.1215**	N-alkyl chain connecting to piperazine ring
2	**C-C**	C17-C18	`NCCN`	**+0.0769**	Piperazine ring C-C bond
3	**C-C**	C27-C28	`NCCN`	**+0.0606**	Piperazine ring C-C bond
4	**C-N**	C15-N16	`CCN(C)C`	**+0.0769**	N-methyl piperazine

Rank	Bond Type	Atoms	Environment	SHAP Value	Chemical Significance
5	C-N	C28-N16	`CCN(C)C`	+0.0702	N-methyl piperazine
6	C-O	C0-O1	`COc`	+0.0672	Methoxy group on phenyl ring

Inactive Compound - Top 5 Key Bonds (Mixed Contribution)

Rank	Bond Type	Atoms	Environment	SHAP Value	Chemical Significance
1	C-S	C1-S2	`CCS(c)(=O)=O`	+0.1120	Methylsulfonyl group (electron-withdrawing)
2	C-S	C22-S30	`csc(c)C`	+0.0982	Thiophene/aromatic sulfur
3	C-C	C9-C10	`cCC(N)=O`	+0.0436	Amide-linked side chain
4	C-N	C12-N13	`CNc(n)s`	+0.0399	Amide linkage to heterocycle
5	C=O	C10=O11	`CC(N)=O`	+0.0280	Carbonyl group (amide)
6	C-C	C7-C8	`ccc(c)C`	+0.0448	Phenyl ring bond
7	C=N	C13-N14	`cnc(N)s`	+0.0260	Thiophene ring bond
8	S=O	S2=O3	`CS(c)(=O)=O`	+0.0216	Sulfonyl group (strong EWG)

🧠 Chemical Interpretation & Key Differences

1. Structural Motifs Driving Activity

Feature	Active Compound	Inactive Compound	Impact
Piperazine Ring	✅ Present (NCCN)	❌ Absent	**Positive driver**
N-Methyl Groups	✅ Multiple N-CH_3	❌ Absent	**Positive driver**
Methoxy Group	✅ OCH_3 on phenyl	❌ Absent	**Positive driver**
Methylsulfonyl	❌ Absent	✅ SO_2CH_3	**Negative driver**
Thiophene Ring	❌ Absent	✅ Present	**Negative driver**
Amide Linkage	❌ Absent	✅ C(=O)N	**Neutral/Negative**

2. Key Chemical Insights

🟢 What Makes the Active Compound Active:

- **Piperazine core** with N-methyl substitutions provides optimal basicity and hydrogen bonding capacity
- **Methoxy-phenyl** group enhances lipophilicity and π-π stacking interactions
- **Chlorophenyl** substituent adds hydrophobic interactions

- **Balanced polarity** from multiple nitrogen atoms enables target binding

🔴 What Makes the Inactive Compound Inactive:

- **Methylsulfonyl group (SO_2CH_3)** is strongly electron-withdrawing, reducing basicity
- **Thiophene aromatic system** may cause unfavorable steric or electronic effects
- **Amide linkage** introduces rigidity that may prevent optimal binding conformation
- **Higher lipophilicity** from phenyl rings may reduce solubility/bioavailability

3. EdgeSHAPer Trustworthiness Comparison

Metric	Active	Inactive	Interpretation
Trustworthiness	0.8842	0.8887	Both explanations are highly reliable (>0.85)
Fidelity	0.8697	0.8068	Active compound explanation is more faithful to model
Infidelity	0.1008	0.0110	Inactive has lower error in removed edges

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📊 Summary Table: Active vs Inactive

Aspect	Active Compound	Inactive Compound
Key Structural Feature	Piperazine with N-methyl	Methylsulfonyl + Thiophene
Dominant SHAP Signal	Positive (N-C, C-C bonds)	Mixed (C-S, S=O bonds)
Electronic Effect	Electron-donating groups	Electron-withdrawing groups
Lipophilicity	Moderate (balanced)	High (phenyl-rich)
Binding Potential	High (flexible, basic)	Low (rigid, acidic)

🎯 Conclusions

1. **The GNN model learned meaningful chemical patterns** - it correctly identifies the importance of piperazine nitrogen atoms and methoxy groups for activity
2. **EdgeSHAPer successfully differentiated** the two compounds by highlighting:
   - **Active**: Piperazine ring bonds (N-C, C-C) as key contributors
   - **Inactive**: Methylsulfonyl (C-S, S=O) and thiophene bonds as key contributors
3. **The model's decision boundary** appears to be driven by:
   - ✅ Basic nitrogen atoms (piperazine) → **Activity**
   - ✅ Electron-donating groups (methoxy) → **Activity**

- ❌ Electron-withdrawing groups (sulfonyl) → **Inactivity**
- ❌ Rigid aromatic systems (thiophene) → **Inactivity**

4. **Trustworthiness scores >0.88** for both compounds indicate the explanations are **reliable and consistent** with the model's internal representations

This analysis demonstrates that the GCN model has learned **chemically interpretable** features for compound activity prediction! 🧪✨